☐ High Priority

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner:	County of Hawaii		
Contractor:	Nan, Inc.	**Project No.:**	WW-4705R
Project Name:	Hilo WWTP Phase 1	**Submittal Number:**	15958-002.1
Submittal Title:	Vibration Monitoring Mechanical Equipment Testing	☐ For Information Only	
TO:	County		
From:	Nan Inc.		

Specification No. and Subject of Submittal / Equipment Supplier			
Spec: 15958	**Paragraph:** 3.07(e)		
Authored By: Pdm Specialist		**Date Submitted:** 09/12/2025	

Submittal Certification

Check Either (A) or (B):

☐ (A) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with no exceptions.

☑ (B) We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings except for the deviations listed.

Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.

General Contractor's Reviewer's Signature: Matthew Chun

Printed Name and Title: Matthew Chun, QC Engineer

In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.

Firm: **Signature:** **Date Returned:**

PM/CM Office Use

Date Received GC to PM/CM:

Date Received PM/CM to Reviewer:

Date Received Reviewer to PM/CM:

Date Sent PM/CM to GC:

Nan, Inc

PROJECT: HILO WWTP REHABILITATION AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY THIS CONTRACTOR. IT IS CERTIFIED CORRECT, COMPLETE, AND IN COMPLIANCE WITH CONTRACT DRAWINGS AND SPECIFICATIONS. ALL AFFECTED CONTRACTORS AND SUPPLIERS ARE AWARE OF, AND WILL INTEGRATE THIS SUBMITTAL (UPON APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED 9/12/2025
SPECIFICATION SECTION # 15958
SPECIFICATION mECHANICAL Equipt & Test
PARAGRAPH 3.07(e)
DRAWING n/a
SUBCONTRACTOR n/a
SUPPLIER n/a
MANUFACTURER n/a
CERTIFIED BY CQCM or Designee : Mark McCarthy

Review Comments:

1. Include vendor qualifications within the submitted file, not as a separate download.

Incorporated - see page 25

Nan Inc

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Hilo Wastewater Treatment Plant Phase 1
Contract No. WW-4705R

VARIANCE

Request Form

Variance Request ____

The decision to grant or deny a variance is at the sole discretion of the relevant authority or entity responsible for reviewing such requests.

SKF Vibration Analyzers and Sensors

1. What are the benefits the variance has to the Government?

Better Testing Capabilities See Item 2.

2. What are the positive and negative impacts to the project and Government if the variance is accepted?

Better Testing Capabilities - Level 2 testing is typically used for VFD driven equipment and takes a bit more time. One of the requirements, “natural frequency testing (3.08.b.2.c)” can be done with a hammer, but we prefer a more thorough method: running the equipment from minimum to maximum speed to identify any critical frequencies. If we detect a critical speed that excites a component’s natural frequency, we can run additional tests to confirm and assess whether it’s a concern.

IIn section 3.0.7 E.1.2 Entek no longer exists for instance, and the CSI 2120 E.2.1 is hardware developed around 2000 and is no longer supported as it has been replaced by the 2140 which is about 10yrs old itself. We have much better resolution A/D converters, etc and the dBX90 from SKF was just released last year.

3. Discuss how the variance is “equal to” or “better than” the specification requirement.

PDM have supported projects as subcontractors for Hawaii Engineering, Hensel Phelps, and others, meeting similar FAT testing requirements using SKF Vibration Analyzers and Sensors. These are at least equivalent to the analyzers listed in the specs you provided, and more than sufficient for the testing needed.

4. What is the cost of the product/material that is originally specified?

None Noted

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List of Variation[s]

Note: these numbers correspond to the spec section

1. SKF Microlog Analyzer dBX in lieu of Rockwell and CSI Analyzers

SECTION 15958

MECHANICAL EQUIPMENT TESTING

PART 1 GENERAL

1.01 SUMMARY

A. Section includes: Testing of mechanical equipment and systems.

1.02 REFERENCES

A. American National Standards Institute (ANSI):
   1. S1.4 Specification for Sound Level Meters.

B. Hydraulic Institute (HI).

C. National Institute of Standards and Technology (NIST).

1.03 SUBMITTALS

A. Provide Source Test Plans as specified in Section 01756 - Commissioning.

B. Provide Installation and Functional Testing Plans as specified in Section 01756 - Commissioning.

C. Schedule of source (factory) tests, Owner staff training, installation testing, pretest checks, clean water facility testing, functional testing, performance testing, start-up testing, and closeout documentation, as specified in this Section and in Section 01756 - Commissioning.

D. Provide vendor operation and maintenance manual as specified in Section 01782 - Operation and Maintenance Manuals.
   1. Include motor rotor bar pass frequencies for motors larger than 500 horsepower.

E. Test instrumentation calibration data.

F. Test reports as specified in this Section and in Section 01756 - Commissioning and equipment sections.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.01 GENERAL

A. Commissioning of equipment as specified in:
   1. This Section.
   2. Section 01756 - Commissioning.
   3. Equipment sections:
      a. If testing requirements are not specified, provide Level 1 Tests.

B. Comply with latest version of applicable standards.

C. Test and prepare piping as specified in Section 15956 - Piping Systems Testing.

D. Operation of related existing equipment:
   1. Owner will operate related existing equipment or facilities necessary to accomplish the testing.
   2. Schedule and coordinate testing as required by Section 01756 - Commissioning.

E. Provide necessary test instrumentation that has been calibrated within 1 year from date of test to recognized test standards traceable to the NIST or approved source:
   1. Properly calibrated field instrumentation permanently installed as a part of the Work may be utilized for tests.
   2. Prior to testing, provide signed and dated certificates of calibration for test instrumentation and equipment.

F. Test measurement and result accuracy:
   1. Use test instruments with accuracies as recommended in the appropriate referenced standards. When no accuracy is recommended in the referenced standard, use 1 percent or better accuracy test instruments:
      a. Improved (lower error tolerance) accuracies specified elsewhere prevail over this general requirement.
   2. Do not adjust results of tests for instrumentation accuracy:
      a. Measured values and values directly calculated from measured values shall be the basis for comparing actual equipment performance to specified requirements.

G. Report features:
   1. Report results in a bound document in generally accepted engineering format with title page, written summary of results compared to specified requirements, and appropriate curves or plots of significant variables in English units.
   2. Include appendix with a copy of raw, unmodified test data sheets indicating test value, date and time of reading, and initials of person taking the data.
   3. Include appendix with sample calculations for adjustments to raw test data and for calculated results.
   4. Include appendix with the make, model, and last calibration date of instrumentation used for test measurements.
   5. Include in body of report a drawing or sketch of the test system layout showing location and orientation of the test instruments relative to the tested equipment features.

3.02 VARIABLE SPEED EQUIPMENT TESTS

A. Establish performance over the entire speed range and at the average operating condition.

B. Establish performance curves for the following speeds:
   1. The speed corresponding to the rated maximum capacity.
   2. The speed corresponding to the minimum capacity.
   3. The speed corresponding to the average operating conditions.

3.03 PUMP TESTS, ALL LEVELS OF TESTING

A. Test in accordance with the following:
   1. Applicable HI Standards.
   2. This Section.
   3. Equipment sections.

B. Test tolerances: In accordance with appropriate HI Standards, except the following modified tolerances apply:
   1. From 0 to plus 5 percent of head at the rated design point flow.
   2. From 0 to plus 5 percent of flow at the rated design point head.
   3. No tolerance for head and flow when ranges are specified.
   4. No negative tolerance for the efficiency at the rated design point, and other specified conditions.
   5. Use of specified test tolerances shall not result in motor overload while operating at any point on the supplied pump operating head-flow curve, including runout.
   6. No positive tolerance for vibration limits. Vibration limits and test methods in HI Standards do not apply, use limits and methods specified in this or other Sections of the Specifications.

3.04 DRIVERS TESTS

A. Test motors as specified in Section 16222 - Low Voltage Motors up to 500 Horsepower.

B. Test other drivers as specified in the equipment section.

3.05 NOISE REQUIREMENTS AND CONTROL

A. Perform noise tests in conjunction with vibration test analysis.

B. Make measurements in relation to reference pressure of 0.0002 microbar.

C. Make measurements of emitted noise levels on sound level meter meeting or exceeding ANSI S1.4, Type II.

D. Set sound level meter to slow response.

E. Unless otherwise specified, maximum free field noise level not to exceed 85 dBA measured as sound pressure level at 3 feet from the equipment.

3.06 PRESSURE TESTING

A. Hydrostatically pressure test pressure containing parts at the appropriate standard or code required level above the equipment component specified design pressure or operating pressure, whichever is higher.

3.07 INSPECTION AND BALANCING

A. Statically and dynamically balance each of the individual rotating parts as required to achieve the required field vibration limits.

B. Statically and dynamically balance the completed equipment rotating assembly and drive shaft components.

C. Furnish copies of material and component inspection reports including balancing reports for equipment system components and for the completed rotating assembly.

D. Critical speed of rotating equipment:
   1. Satisfy the following:
      a. The first lateral and torsional critical speed of all constant, variable, and 2-speed driven equipment that is considered rigid such as horizontal pumps, all non-clog pumps, blowers, air compressors, and engines shall be at least 25 percent above the equipment™s maximum operating speed.
      b. The first lateral and torsional critical speed of all constant, variable, and 2-speed driven equipment that is considered flexible or flexibly mounted such as vertical pumps (vertical in-line and vertical non-clog pumps excluded) and fans shall at least 25 percent below the equipment™s lowest operating speed.
      c. The second lateral and torsional critical speed of all constant, variable, and 2-speed equipment that is considered flexible or flexibly mounted shall be at least 25 percent above the maximum operating speed.

E. Vibration tests:
   1. Definitions:
      a. Root mean square: for pumps operating at speeds greater than 600 rpm, the vibration measurement shall be measured as the overall velocity in inches per second root mean square (RMS).
      b. Peak-to-peak displacement: The root means squared average of the peak-to-peak displacement multiplied by the square root of 2.
      c. Peak velocity: The root mean squared average of the peak velocity multiplied by the square root of 2.
      d. Peak acceleration: The root mean squared average of the p[eak] acceleration multiplied by the square root of 2.

equipment does not exist see item 2

      e. High frequency enveloping: A process to extract very low amplitude time domain signals associated with impact or impulse events such as bearing or gear tooth defects and display them in a frequency spectrum of acceleration versus frequency.
1   1) Manufacturers: One of the following or equal:
            a) ~~Rockwell Automation, Entek Group, "Spike Energy" analysis.~~
            b) CSI, "PeakVue."
      f. Rotor bar pass frequency (RBF), for detecting loose rotor bars.

g. Low speed equipment: Equipment or components of equipment rotating at less than 600 revolutions per minute.
h. High speed equipment: Equipment and equipment components operating at or above 600 revolutions per minute.
i. Preferred operating range: Manufacturer™s defined preferred operating range (POR) for the equipment.
j. Allowable operating range: Manufacturer™s defined allowable operating range (AOR) for the equipment.

2. Vibration instrumentation requirements:
   a. Analyzers: Use digital type analyzers or data collectors with anti-aliasing filter, 12-bit A/D converter, fast Fourier transform circuitry, phase measurement capability, time wave form data storage, high-frequency enveloping capabilities, 35 frequency ranges from 21 to 1,500,000 cycles per minute, adjustable fast Fourier transform resolution from 400 to 6,400 lines, storage for up to one hundred 3,200 line frequency spectra, data output port, circuitry for integration of acceleration data to velocity or double integration to displacement.
      1) Manufacturers: One of the following or equal:
         a) Computational Systems Inc., (CSI) Division of Emerson Process Management, Model 2120A, Data Collector/analyzer with applicable analysis software.
         b) Pruftechnik, VIBXPERT II.
   b. Analyzer settings:
      1) Units: English, inches/second, mils, and gravitational forces.
      2) Fast Fourier transform lines: Most equipment 1,600 minimum; for motors, enough lines as required to distinguish motor current frequencies from rotational frequencies, use 3,200 lines for motors with a nominal speed of 3,600 revolutions per minute; 3,200 lines minimum for High Frequency Enveloping; 1,600 lines minimum for low speed equipment.
      3) Sample averages: 4 minimum.
      4) Maximum frequency (Fmax): 40 times rotational frequency for rolling element bearings, 10 times rotational frequency for sleeve bearings.
      5) Amplitude range: Auto select but full scale not more than twice the acceptance criteria or the highest peak, whichever is lower.
      6) Fast Fourier transform windowing: Hanning Window.
      7) High pass filter: Minus 3 dB at 120 cycles per minute for high speed equipment. Minus 3 dB at 21 cycles per minute for low speed equipment.
   c. Accelerometers:
      1) For low speed equipment: Low frequency, shear mode accelerometer, 500 millivolts per gravitational force sensitivity, 10 gravitational force range, plus/minus 5 percent frequency response from 0.5 hertz to 850 hertz, magnetic mount:
         a) Manufacturers: One of the following or equal:
            (1) Wilcoxon Research, Model 797L.
            (2) PCB, Model 393C.

2) For high speed equipment: General purpose accelerometer, 100 millivolts per gravitational force sensitivity, 50 gravitational force range, plus/minus 3dB frequency response range from 2 hertz to 12,000 hertz when stud mounted, with magnetic mount holder:
    a) Manufacturers: One of the following or equal:
        (1) Wilcoxon Research, Model 793.
        (2) Entek-IRD Model 943.

3. Accelerometer mounting:
    a. Use magnetic mounting or stud mounting.
    b. Mount on bearing housing in location with best available direct path to bearing and shaft vibration.
    c. Remove paint and mount transducer on flat metal surface or epoxy mount for High Frequency Enveloping measurements.

4. Vibration acceptance criteria:
    a. Testing of rotating mechanical equipment: Tests are to be performed by an experienced, factory trained, and independent authorized vibration analysis expert.
    b. Vibration displacement limits: Unless otherwise specified, equipment operating at speeds 600 revolutions per minute or less is not to exhibit unfiltered readings in excess of following:

Operating Conditions and Application Data	Overall Peak- to-Peak Displacement	
	Field, mils	Factory, mils
Operation within the POR	3.0	4.0
Operation within the AOR	4.0	5.0
Additive value when measurement location is greater than 5 feet above foundation.	2.0	2.0
Additive value for solids-handling pumps	2.0	N/A
Additive value for slurry pumps	2.0	N/A

    c. Vibration velocity limits: Unless otherwise specified, equipment operating at speeds greater than 600 revolutions per minute is not to exceed the following peak velocity limits:

HI Pump Type	Horsepower	Field Test	Factory Test
		Overall RMS	Overall RMS
Horizontal Solids Handling Centrifugal Pumps	Below 33 hp	0.25	0.28
Horizontal and Vertical In-Line Centrifugal Pumps (other than non-clog type)	Between 33 and 100 hp	0.28	0.31
	100 hp and above	0.31	0.34
Vertical Solids Handling Centrifugal Pumps	Below 33 hp	0.30	0.33

HI Pump Type	Horsepower	Field Test Overall RMS	Factory Test Overall RMS
Vertical Turbine, Mixed Flow, and Propeller Pumps (solids-handling type pumps)	Between 33 and 100 hp	0.32	0.35
	100 hp and above	0.34	0.35
Non-Solids Handling Centrifugal Pumps HI Types BB1, BB2, BB3, BB4, BB5, OH1, OH2, OH3, OH4, OH5, and OH7	Below 268 hp	0.15	0.19
	268 hp and above	0.19	0.22
Vertical Turbine, Mixed Flow, and Propeller Pumps HI Types VS1, VS2, VS3, VS4, VS5, VS6, VS7, and VS8	Below 268 hp	0.13	
	268 hp and above	0.17	
Slurry Pumps		0.25	0.30
Motors		See Applicable Motor Specification	See Applicable Motor Specification
Gear Reducers, Radial		Not to exceed AGMA 6000-B96 limits	Not to exceed AGMA 6000-B96 limits
Other Reducers, Axial		0.1	N/A

d. Equipment operation: Measurements are to be obtained with equipment installed and operating within capacity ranges specified and without duplicate equipment running.

e. Additional criteria:
   1) No narrow band spectral vibration amplitude components, whether sub-rotational, higher harmonic, or synchronous multiple of running speed, are to exceed 40 percent of synchronous vibration amplitude component without manufacturer's detailed verification of origin and ultimate effect of such excitation.
   2) The presence of discernable vibration amplitude peaks in Test Level 2 or 3 vibration spectra at bearing inner or outer race frequencies shall be cause for rejection of the equipment.
   3) For motors, the following shall be cause for rejection:
      a) Stator eccentricity evidenced by a spectral peak at 2 times electrical line frequency that is more than 40 percent of the peak at rotational frequency.
      b) Rotor eccentricity evidenced by a spectral peak at 2 times electrical line frequency with spectra side bands at the pole pass frequency around the 2 times line frequency peak.

c) Other rotor problems evidenced by pole pass frequency side bands around operating speed harmonic peaks or 2 times line frequency side bands around rotor bar pass frequency or around 2 times the rotor bar pass frequency.

d) Phasing problems evidenced by 1/3 line frequency side band spectral peaks around the 2 times electrical line frequency peak.

4) The presence of peaks in a High Frequency Enveloping spectra plot corresponding to bearing, gear or motor rotor bar frequencies or harmonics of these frequencies shall be cause for rejection of the equipment; since inadequate lubrication of some equipment may be a cause of these peaks, lubrication shall be checked, corrected as necessary and the high frequency envelope analysis repeated.

5. Vibration testing results presentation:

a. Provide equipment drawing with location and orientation of measurement points indicated.

b. For each vibration measurement take and include appropriate data on equipment operating conditions at the time vibration data is taken; for pumps, compressors, and blowers record suction pressure, discharge pressure, and flow.

c. When Vibration Spectra Data required:

1) Plot peak vibration velocity versus frequency in cycles per minute.

2) Label plots showing actual shaft or part rotation frequency, bearing inner and outer race ball pass frequencies, gear mesh frequencies and relevant equipment excitation frequencies on the plot; label probable cause of vibration peaks whether in excess of specification limits or not.

3) Label plots with equipment identification and operating conditions such as tag number, capacity, pressure, driver horsepower, and point of vibration measurement.

4) Plot motor spectra on a log amplitude scale versus frequency.

d. For low speed equipment, plot peak vibration displacement versus frequency as well as velocity versus frequency.

e. Provide name of manufacturer and model number of the vibration instrumentation used, including analyzer and accelerometer used together with mounting type.

3.08 TESTING LEVELS

A. Level 1 Tests:

1. Level 1 Performance Test:

a. General:

1) For equipment, operate, rotate, or otherwise functionally test for 15 minutes minimum after components reach normal operating temperatures.

2) Operate at rated design load conditions.

3) Confirm that equipment is properly assembled.

4) Confirm the equipment moves or rotates in the proper direction.

5) Confirm shafting, drive elements, and bearings are installed and lubricated in accordance with proper tolerances.
6) Confirm that no unusual power consumption, lubrication temperatures, bearing temperatures, or other conditions are observed.

b. Pumps:
1) Comply with general performance test requirements as specified in this Section.
2) Measure flow and head while operating at or near the rated condition; for factory testing, testing may be at reduced speeds with flow and head corresponding to the rated condition when adjusted for speed using the appropriate affinity laws:
a) Use of a test driver is permitted for factory tests when actual driver is given a separate test at its point of manufacture as specified in Section 16222 - Low Voltage Motors up to 500 Horsepower or the applicable equipment section.
b) Use actual driver for field tests.
3) Record measured flow, suction pressure, discharge pressure, and make observations on bearing temperatures and noise levels.

2. Level 1 Vibration Test:
a. Test requirement:
1) Measure filtered vibration spectra versus frequency in 3 perpendicular planes at each normally accessible bearing housing on the driven equipment, any gears and on the driver; 1 plane of measurement to be parallel to the axis of rotation of the component.
2) Vibration spectra versus frequency shall be in accordance with Vibration Acceptance Criteria.

b. Equipment operating condition: Test at specified maximum speed.

3. Level 1 Noise Test:
a. Measure unfiltered overall A-weighted sound pressure level in dBA at 3 feet horizontally from the surface of the equipment and at a mid-point of the equipment height.

B. Level 2 Tests:
1. Level 2 Performance Test:
a. General:
1) For equipment, operate, rotate, or otherwise functionally test for at least 2 hours after components reach normal operating temperatures.
2) Operate at rated design load conditions.
3) Confirm that equipment is properly assembled.
4) Confirm the equipment moves or rotates in the proper direction.
5) Confirm shafting, drive elements, and bearings are installed and lubricated in accordance with proper tolerances.
6) Confirm that no unusual power consumption, lubrication temperatures, bearing temperatures, or other conditions are observed.

b. Pumps:
1) Comply with general performance test requirements as specified in this Section.

2) Test 2 hours minimum for flow and head at the rated condition; for factory testing, testing may be at a reduced speeds with flow and head corresponding to the rated condition when adjusted for speed using the appropriate affinity laws:
    a) Use of a test driver is permitted for factory tests when actual driver is given a separate test at its point of manufacture as specified in Section 16222 - Low Voltage Motors up to 500 Horsepower.
    b) Use actual driver for field tests.
3) Test for flow and head at 2 additional conditions; 1 at 25 percent below the rated flow and 1 at 10 percent above the rated flow.
4) Record measured flow, suction pressure, discharge pressure, and observations on bearing temperatures and noise levels at each condition.

2. Level 2 Vibration Test:
    a. Test requirement:
        1) Measure filtered vibration spectra versus frequency and measure vibration phase in 3 perpendicular planes at each normally accessible bearing housing on the driven equipment, any gears and on the driver; 1 plane of measurement to be parallel to the axis of rotation of the component; measure actual rotational speeds for each vibration spectra measured using photometric or other tachometer input connected directly to the vibration data collector.
        2) Vibration spectra versus frequency shall be in accordance with Vibration Acceptance Criteria.
    b. Equipment operating condition: Repeat test requirements at design specified maximum speed and at minimum speed for variable speed equipment.
    c. Natural frequency test of field installed equipment:
        1) Excite the installed equipment and support system in 3 perpendicular planes, use same planes as operating vibration measurement planes, and determine the as-installed natural resonant frequency of the driven equipment, the driver, gears, and supports.
        2) Perform test at each bearing housing, at each support pedestal, and for pumps on the suction and discharge piping.
        3) Perform with equipment and attached piping full of intended service or process fluid.
3. Level 2 Noise Test:
    a. Measure filtered A-weighted overall sound pressure level in dBA for each of 8 octave band mid-points beginning at 63 hertz measured at 3 feet horizontally from the surface of the equipment at mid-point height of the noise source.

C. Level 3 Tests:
    1. Level 3 Performance Tests:
        a. General:
            1) For equipment, operate, rotate, or otherwise functionally test for at least 4 hours after components reach normal operating temperatures.

2) Operate at rated design load conditions for 1/2 the specified time; operate at each of any other specified conditions for a proportionate share of the remaining test time.
3) Confirm that equipment is properly assembled.
4) Confirm the equipment moves or rotates in the proper direction.
5) Confirm shafting, drive elements, and bearings are installed and lubricated in accordance with proper tolerances.
6) Confirm that no unusual power consumption, lubrication temperatures, bearing temperatures, or other conditions are observed.
7) Take appropriate capacity, power or fuel consumption, torque, revolutions per minute, pressure, and temperature readings using appropriate test instrumentation to confirm equipment meets specified performance requirements at the design rated condition.
8) Bearing temperatures: During maximum speed or capacity performance testing, measure and record the exterior surface temperature of each bearing versus time.

b. Pumps:
1) Comply with general performance test requirements as specified in this Section.
2) Test 4 hours minimum for flow and head; begin tests at or near the rated condition; for factory and field-testing, test at full speed.
   a) Use of a test driver is permitted for factory tests when actual driver is given a separate test at its point of manufacture as specified in Section 16222 - Low Voltage Motors up to 500 Horsepower.
   b) Use actual driver for field tests.
3) Test each specified flow and head condition at the rated speed and test at minimum as well as maximum specified speeds; operate at each test condition for a minimum of 15 minutes or longer as necessary to measure required performance, vibration, and noise data at each test condition.
4) Record measured shaft revolutions per minute, flow, suction pressure, discharge pressure; record measured bearing temperatures (bearing housing exterior surface temperatures may be recorded when bearing temperature devices are not required by the equipment section) and record observations on noise levels.
5) Perform efficiency and/or Net Positive Suction Head Required (NPSHr) and/or priming time tests when specified in the equipment section in accordance with the appropriate HI standard and as follows:
   a) Perform NPSHr testing at maximum rated design speed, head and flow with test fluids at ambient conditions; at maximum rated speed, test at 15 percent above rated design flow, and 25 percent below rated design flow.
   b) Perform efficiency testing with test fluids at maximum rated speed.
   c) Perform priming time testing with test fluids at maximum rated speed.

2. Level 3 Vibration Test:
   a. Requirements: Same as Level 2 vibration test except data taken at each operating condition tested and with additional requirements below.
   b. Perform High Frequency Enveloping Analysis for gears and bearings.
      1) Measure bearing element vibration directly on each bearing cap in a location close as possible to the bearing load zone that provides a smooth surface and direct path to the bearing to detect bearing defects.
      2) Report results in units of acceleration versus frequency in cycles per minute.
   c. Perform Time Wave Form analysis for gears, low speed equipment and reciprocating equipment; plot true peak amplitude velocity and displacement versus time and label the period between peaks with the likely cause of the periodic peaks (relate the period to a cause).
   d. Plot vibration spectra on 3 different plots; peak displacement versus frequency, peak acceleration versus frequency and peak velocity versus frequency.
3. Level 3 Noise Test: Measure filtered, un-weighted overall sound pressure level in dB at 3 feet horizontally from the surface of the equipment at mid-point height and at 4 locations approximately 90 degrees apart in plain view; report results for each of 8 octave band mid-points beginning at 63 hertz.

D. Level 4 Tests:
1. Level 4 Performance Test:
   a. General:
      1) For equipment, operate, rotate, or otherwise functionally test for at least 8 hours after components reach normal operating temperatures.
      2) Operate at rated design load conditions for 1/2 the specified time; operate at each of any other specified conditions for a proportionate share of the remaining test time.
      3) Confirm that equipment is properly assembled.
      4) Confirm the equipment moves or rotates in the proper direction.
      5) Confirm shafting, drive elements, and bearings are installed and lubricated in accordance with proper tolerances.
      6) Confirm that no unusual power consumption, lubrication temperatures, bearing temperatures, or other conditions are observed.
      7) Take appropriate capacity, power or fuel consumption, torque, revolutions per minute, pressure and temperature readings, using appropriate test instrumentation to confirm equipment meets specified performance requirements at the design rated condition.
      8) Bearing temperatures: During maximum speed or capacity testing, measure and record the exterior surface temperature of each bearing versus time.
   b. Pumps:
      1) Comply with general performance test requirements as specified in this Section.
      2) Test 8 hours minimum for flow and head; begin tests at or near the rated condition; for factory and field-testing, test with furnished motor at full speed.

3) Test each specified flow and head condition at the rated speed and test at minimum as well as maximum specified speeds; operate at each test condition for a minimum of 20 minutes or longer as necessary to measure required performance, vibration, and noise data at each test condition.
4) Record measured shaft revolutions per minute, flow, suction pressure, discharge pressure; record measured bearing temperatures (bearing housing exterior surface temperatures may be recorded when bearing temperature devices not required by the equipment section) and record observations on noise levels.
5) Bearing temperatures: During maximum speed or capacity testing, measure and record the exterior surface temperature of each bearing versus time.
6) Perform efficiency and/or Net Positive Suction Head Required (NPSHr) and/or priming time tests when specified in the equipment section in accordance with the appropriate HI standard and as follows:
    a) Perform NPSHr testing at maximum rated design speed, head, and flow with test fluids at ambient conditions; at maximum rated speed, test at 15 percent above rated design flow, and 25 percent below rated design flow.
    b) Perform efficiency testing with test fluids at maximum rated speed.
    c) Perform priming time testing with test fluids at maximum rated speed.

2. Level 4 Vibration Test: Same as Level 3 vibration test.
3. Level 4 Noise Test: Same as Level 3 Noise Test except with data taken at each operating condition tested.

3.09 PLANNING PHASE

A. Submit test plans as specified in Section 01756 - Commissioning:
1. Indicate test start time and duration, equipment to be tested, other equipment involved or required; temporary facilities required, number and skill or trade of personnel involved; safety issues and planned safety contingencies; anticipated effect on Owner's existing equipment and other information relevant to the test.
2. Provide locations of all instruments to be used for testing. Provide calibration records for all instrumentation.

3.10 COMMISSIONING PHASE

A. Source testing:
1. Witnessing not required unless specified otherwise in equipment section.
2. Witnessed tests: Schedule test date and notify Engineer at least 30 days prior to start of test.
3. Test equipment as specified in Section 01756 - Commissioning.
4. Test fluids as specified in Section 01756 - Commissioning.
5. Submit reports as specified in Section 01756 - Commissioning.

B. Installation testing:
   1. Test equipment as specified in Section 01756 - Commissioning.

C. Clean Water Facility Testing:
   1. Test equipment as specified in Section 01756 - Commissioning.

D. Functional testing(Performance testing):
   1. Witnessing required as specified in Section 01756 - Commissioning.
   2. Schedule test date and notify Engineer at least 7 days prior to start of test.
   3. Test equipment as specified in equipment sections. Test fluids as specified in Section 01756 - Commissioning.
   4. Submit reports as specified in Section 01756 - Commissioning.
   5. Test fluids as specified in Section 01756 - Commissioning.

E. Closeout documentation:
   1. Provide closeout documentation as specified in Section 01756 - Commissioning and equipment sections.

3.11 PRE-START-UP AND STARTUP PHASE

A. As specified in Section 01756 - Commissioning and equipment sections.

END OF SECTION

Qualifications

Scott Mercer,brings over 20 years of experience as a senior analyst and instructor with PdM, and he holds the highest certifications in the field; including a current ISO Category 4 certification (attached). He™s performed similar tests for Hawaii County Department of Water Supply, Oahu™s Board of Water Supply, and various contractors across the islands, including work at Kailua WWTP and Sand Island.

We™ve also supported projects as subcontractors for Hawaii Engineering, Hensel Phelps, and others, meeting similar FAT testing requirements using SKF Vibration Analyzers and Sensors. These are at least equivalent to the analyzers listed in the specs you provided, and more than sufficient for the testing needed

Technical Associates

TABofC

Board of Certification

Certifies That

Scott D. Mercer

has successfully completed the requirements for

VIBRATION ANALYST: ISO CATEGORY IV

TABofC Certification Program for Vibration Analysts complies with the International Organization for Standardization, ISO 18436-2 Standard, Condition Monitoring and Diagnostics of Machines.

October 11, 2018	13/10/11 7800-735-04	October 11, 2023
Date of Renewal	*Certificate Number*	*Expiration Date*

Technical Associates Board of Certification
1230 West Morehead Street, Suite 400
Charlotte, NC 28208
Tel: (704) 333-9011 Fax: (704) 333-1728

James E. Berry, P.E.

James E. Berry, P.E., President

Certification Authority

Technical Associates Board of Certification (TABofC) is a certifying body that adheres to the following:

ISO/IEC 17024:2012 *Conformity assessment – General requirements for bodies operating certification of persons.*

ISO 18436-1:2012 *Condition monitoring and diagnostics of machines – Requirements for qualification and assessment of personnel Part 1: Requirements for assessment bodies and the assessment process.*

Evidence of Certification

TABofC conducts certification assessment(s) of an individuals' expertise in condition monitoring and diagnostics of machines utilizing, and adhering to, the specific requirements found in:

ISO18436-2:2014 *Condition monitoring and diagnostics of machines – Requirements for qualification and assessment of personnel Part 2: Vibration condition monitoring and diagnostics*

TABofC attests to the qualifications of the individual to whom this certificate is given but does not give authority to perform condition monitoring and diagnostics of machinery. It is understood that it is the responsibility of the employer to authorize the certificate-holder to perform these tasks and take responsibility for any/all results.

Conditions of Certification

This certificate is property of TABofC and is valid only for the stated range of dates, and only if the certificate-holder is emotionally and physically capable of performing their duties and adheres to the Code of Ethics.

Certification Renewal Information

You are responsible for keeping Technical Associates Board of Certification updated with your current contact information. Please notify TABofC of any changes to your email, home address, phone number or place of employment.

Per ISO 18436-1:2012, your certification is valid for 5 years and will expire if it is not renewed during the 6-month period prior to the expiration date shown on your certificate.

TECHNICAL ASSOCIATES
BOARD OF CERTIFICATION

TABofC

1230 WEST MOREHEAD STREET, SUITE 400 CHARLOTTE, NC 28208 PHONE: 704-333-9011 FAX: 704-333-1728

August 14, 2018

Scott D. Mercer
14240 SE Lyon Street
Happy Valley, OR 97086

Dear Scott,

Congratulations! We are pleased to inform you that an ISO Category IV Certificate has been prepared for you and is included with this letter. This certification means you have demonstrated that you meet the minimum requirements for certification renewal through ongoing work experience, training, professional development and other related activities. The expiration date of your renewed certification is indicated on your new certificate.

You are responsible for keeping Technical Associates Board of Certification (TABofC) updated with your current contact information. Please notify us of any changes to your personal or business email, home address, cell phone number or place of employment. This information can be submitted on our website under "Contact Information" or sent directly to Tom Hicks (thicks@technicalassociates.net). Updating your contact information will allow TABofC to alert you with advance notification prior to your certification expiration date.

Thank you for your support of the TABofC certification program, and we wish you and your organization continued success. If you have any questions please do not hesitate to contact us directly at (704) 333-9011 or visit our website at www.technicalassociates.net.

Sincerely,

James E. Berry, P.E.

James E. Berry, P.E.
President
Technical Associates of Charlotte, P.C.

TECHNICAL ASSOCIATES
BOARD OF CERTIFICATION

TABofC

1230 WEST MOREHEAD STREET, SUITE 400 CHARLOTTE, NC 28208 PHONE: 704-333-9011 FAX: 704-333-1728

FORM F-20
CERTIFICATION RENEWAL INFORMATION

You are responsible for keeping Technical Associates Board of Certification updated with your current contact information. Please notify TABofC of any changes to your email, home address, phone number or place of employment. This information can be submitted on our web site under "Contact Information" or sent directly to Tom Hicks (thicks@technicalassociates.net). Updating your contact information will allow TABofC to alert you with your advance notification prior to your certification expiration date.

Per ISO 18436-1:2012, your certification is valid for 5 years and will expire if it is not renewed during the 6-month period prior to the expiration date shown on your certificate. Certificate-holders who have not renewed their certification will have their name removed from the TABofC website upon expiration of his/her certificate.

To be eligible for renewal, you will need to submit an application form to TABofC during the 6-month period prior to the expiration date. The Application for Renewal of Certification is available on the TABofC website (www.technicalassociates.net).

In order to qualify for renewal, a minimum number of credits needs to be accrued for the given category (I-30, II-40, III-55, IV-70). Credits can be attained through your Work Experience, Related Activities and Continuing Education. On the application you also must attest that you can provide verifiable evidence of continued satisfactory work activity, without significant interruption, in the vibration analysis field. For additional details, please refer to the Application for Renewal of Certification that is available on the TABofC website (www.technicalassociates.net).

A nominal processing fee will be charged only if you are granted this certification renewal.

Variance - Testing Equipment

SKF

SKF Microlog Analyzer dBX

Handheld data collector and analyzer

Large screen.
Fast measurements.
Easy to use.

Data collection and analysis the way you want it

The SKF Microlog Analyzer dBX has been developed with users in mind. This is why it has a large screen, developed for efficient analysis in the field. This is why it has SKF's most powerful processing power for fast measurements and the ability to perform analyses on the device. This is why it is more user-friendly than ever.

Our largest screen

The SKF Microlog Analyzer dBX has a 10.1-inch high-resolution touch screen. It can display up to 6 measurement windows at once and has been designed for easy viewing in any light condition.

Fast measurements with MPA-in-a-flash

Multi-Point Acquisition (MPA) is SKF's fastest vibration analysis method. It is typically three times faster than the previous Microlog series, saving you time spent on taking measurements. For even faster data collection, the SKF Microlog Analyzer dBX can take simultaneous tri-axial measurements.

Easy to use

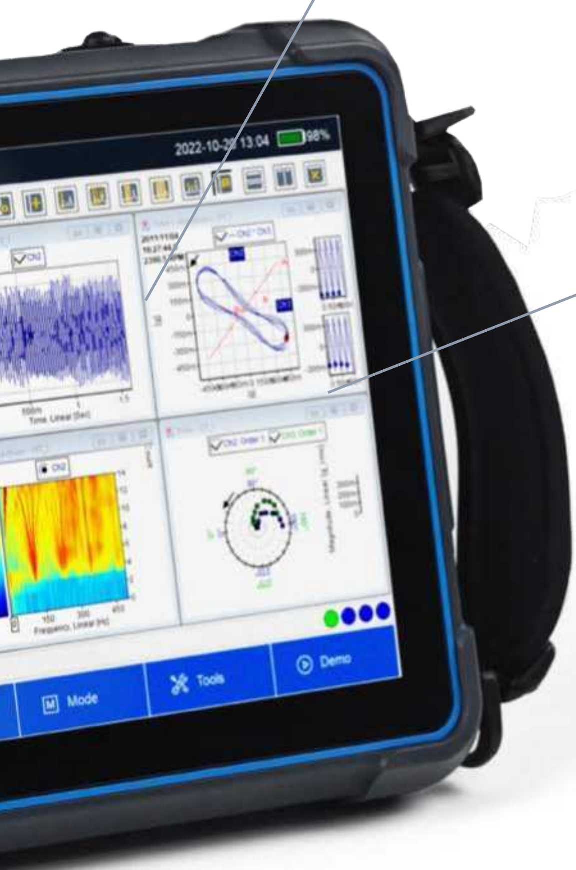
New, improved user interface makes the SKF Microlog Analyzer dBX easier to learn and use than the previous model. Getting you started quickly, with minimum training.

Handle with or without gloves

Physical backlit keypad and touch screen control.

Advanced on-device data analysis

Possibility to perform data analyses on the device, without the need to use an external computer. Equipped with SKF gE enveloping to determine the condition of rolling element bearings.



Large storage memory

256 GB of memory. Enough storage to meet your individual analysis needs.

Designed for field work

Rugged design for reliability in industrial environments. Dust and water-ingress protection to IP65. Onboard camera and RFID reader to assist in identifying rotating machinery location.

Hours of power

Interchangeable and rechargeable battery supports up to 8 hours of continuous data collection.

Ready for standalone or online work

The SKF Microlog Analyzer dBX can be used as a standalone tool, taking measurements and analysing data from rotating machinery to determine machine health. It can also be connected to SKF monitoring software, such as the SKF @ptitude Analyst, to handle analysis, on-premises and remote data storage, and download predefined measurement routes to the dBX.

Standard or extended version

The SKF Microlog Analyzer dBX is available in two versions. The standard version has three main tools – SKF Data Collector, SKF dBX Analyzer and SKF Balancing – all you need for most data collection and fault-tracing tasks.

The extended version has the same tools as the standard version but has four additional tools for more embedded analysis capability. The extra tools are SKF RunUp CoastDown, SKF Order Tracking, SKF Data Recorder and SKF Orbit Analysis.

Standard version
CMVA 90-M-CK-SL

- Data Collector
- dBX Analyzer
- Balancing
- gE Enveloping
- Bump Test
- ODS Analysis

Extended version
CMVA 90-F-CK-SL

- Data Collector
- dBX Analyzer
- Balancing
- gE Enveloping
- Bump Test
- ODS Analysis
- RunUp CoastDown
- Order Tracking
- Data Recorder
- Orbit Analysis

Built-in tools

The SKF Microlog Analyzer dBX hardware is available with a number of valuable software tools.

SKF Data Collector
Collect data based on defined routes, measurement points and sensor set-up.

SKF dBX Analyzer
Collect measurements and perform additional analysis.

SKF Balancing
Balance different types of rotating machines with step-by-step guidance.

SKF gE Enveloping
Determine the condition of rolling element bearings using SKF analysis algorithms.

SKF RunUp CoastDown
Used to search for dangerous resonances during machine start-up or shut-down.

SKF Bump Test
Look for structural resonances in machinery and installations (using a hammer).

SKF ODS Analysis
Determine structural resonances in machinery and installations using Operational Deflection Shape analysis.

SKF Order Tracking
Perform (harmonic) order-based vibration analysis suitable for variable speed machinery.

SKF Data Recorder
Record long real-time measurements and playback to investigate rotating machinery issues.

SKF Orbit Analysis
Visualise rotational behaviour of a rotating shaft. Often used to diagnose faults in machinery with oil-film bearings.

Technical specifications for SKF Microlog Analyzer dBX, CMVA 90

Measurements

Input channels:	4 analogue input channels with IEPE bias voltage, tacho channel with built-in power supply for laser tachometer
Data acquisition:	24-bit A/D converter (>90 dB dynamic range)
Max. bandwidth:	40 kHz (102.4 kHz sampling rate)
Accuracy:	± 2.5% of full scales range
Measurements parameters:	acceleration, velocity, displacement, SKF gE bearing condition, phase, voltage, and speed
Multi-point automation:	Up to 12 measurements can be listed for one button push automated data collection at each measurement location

Environmental

Operating temperature:	-10 to +50 °C *(13 to + 122 °F)*
Storage temperature:	-20 to +60 °C *(-4 to + 140 °F)*
IP rating:	IP65 dust and water ingress to EN 60529 specification
Rugged:	1.2 m *(4 ft)* drop test to MIL STD 810 specification
Approvals:	CE, UKCA, KC, RCM

Physical

Dimensions:	300 x 195 x 50 mm *(11.8 x 7.7 x 1.97 in)*
Weight:	1.9 kg *(4.2 lbs)*, 1.7 kg using single battery
Keypad:	Backlit keys, up, down, right and left, OK, cancel, menu key, right click, cursor toggling, zoom, start/stop measurement key, power on/off
Connectors:	BNC on 4 input channels, 6-pin Fischer, 7-pin Fischer (Microlog CMXA series compatible)
LCD screen:	10.1 in. multi-point colour touch screen, 1280 x 800 pixels, for indoor and outdoor use
Camera:	built-in, rear facing camera
RFID tag reader:	built-in, located on rear
PC interface:	USB A-type connector

Power source

Power:	2x Lithium-ion polymer rechargeable/ swappable batteries
Battery:	up to 8 hours

Ordering information

CMVA 90-M-CK-SL kit includes:

- CMVA 90-M Microlog dBX, programmed with SKF DataCollector, SKF dBX Analyzer and SKF Balancing applications, 256 GB on-board storage
- CMAC 9001 Universal power supply with 4 power cords
- CMAC 9002 Power adapter cable
- CMAC 9005 Two (2) batteries
- CMAC 9010 USB A-Type to A-Type communications cable
- CMAC 9015 SKF branded carry case
- CMAC 9016 Two (2) hand-straps
- CMAC 9017 Neck-strap
- CMSS 2200 General purpose accelerometer, cable and magnet mount
- Certificate of calibration and conformance

CMSS 1500-K sensor kit includes:

- General purpose, 100 mV/g accelerometer
- Accelerometer coiled cable, BNC connector
- Magnet with fixing bolt
- Certificate of calibration

CMVA 90-F-CK-SL kit includes:

- CMVA 90-F Microlog dBX, programmed with SKF DataCollector, SKF dBX Analyzer and SKF Balancing, SKF Order Deflection Tracking and SKF Raw data recorder applications, 256 GB on-board storage,
- CMAC 9001 Universal power supply with 4 power cords
- CMAC 9002 Power adapter cable
- CMAC 9005 Two (2) batteries
- CMAC 9010 USB A-Type to A-Type communications cable
- CMAC 9015 SKF branded carry case
- CMAC 9016 Two (2) hand-straps
- CMAC 9017 Neck-strap
- CMSS 2200 General purpose accelerometer, cable and magnet mount
- Certificate of calibration and conformance

Add the accessories you need

A number of accessories are available to complement the SKF Microlog Analyzer dBX. To allow for maximum application flexibility, the device is not supplied with vibration sensors or cables in the box, but these can be ordered separately.

skf.com/microlog-dbx

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